Final Examination: Machine Learning

I. True/False Evaluation with Justification (20 points)

Evaluate the following statements. For each statement, declare whether it is True or False and provide a concise, theoretically sound justification.

1. Boosting algorithms are exclusively applicable to regression analysis.
2. Linear Discriminant Analysis (LDA) are classified as unsupervised learning algorithms.
3. Uncertainty sampling is the only query strategy utilized within active learning frameworks.
4. Consider two models, A and B. On a given training dataset, Model A achieves an accuracy of 90%, while Model B achieves 99%. It is guaranteed that Model A will yield inferior performance compared to Model B on an unseen test set.

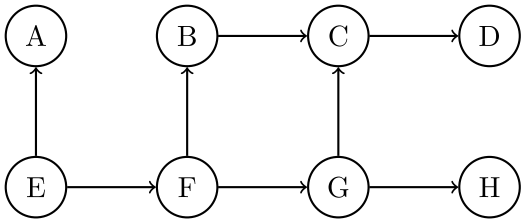
II. Clustering Metrics and Analysis (25 points)

Suppose we have a dataset consisting of 100 samples with ground truth labels distributed as follows: samples 1 through 20 belong to Class 1, samples 21 through 40 belong to Class 2, and samples 41 through 100 belong to Class 3. A clustering algorithm partitions this data into two clusters: Cluster c_1 contains only sample 1, and Cluster c_2 contains samples 2 through 100.

1. Calculate the purity of this clustering assignment. (8 points)
2. What is the theoretical maximum value of purity for any clustering assignment? (2 points)
3. Is it possible to achieve this theoretical maximum purity on this specific dataset? If so, provide an explicit example of such a cluster assignment. (5 points)
4. A researcher claims, "When applying unsupervised learning methodologies, we can incorporate the purity metric directly into the training objective to improve clustering." Analyze whether this statement is fundamentally correct or flawed, and provide a theoretical justification. (5 points)
5. Normalized Mutual Information (NMI) is another common evaluation metric. Is a higher NMI value always indicative of a superior clustering result? Provide a mathematical or conceptual justification. (5 points)

III. Static Bayesian Network Analysis (15 points)

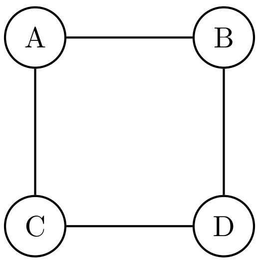
Consider a probabilistic graphical model structured as a Static Bayesian Network, defined by the following Directed Acyclic Graph (DAG):



1. Derive the fully factorized expression for the joint probability distribution based on the topological structure of the provided network.
2. Construct and draw the corresponding factor graph for this network, clearly denoting variable nodes and factor nodes.
3. Using the d-separation criterion, rigorously analyze whether node A is conditionally independent of node C, given that node F is observed.

IV. Markov Random Field Mechanics (20 points)

Consider the following undirected graphical model (Markov Random Field):



1. Identify and comprehensively list the set of all maximal cliques within this undirected graph.
2. Draw the corresponding factor graph for this Markov Random Field.
3. Write a potential function $\phi_C(X_C)$ that assigns non-negative values to each maximal clique C. Then, given a specific realization (a, b, c, d) for the variables (A, B, C, D), write down the complete factorized expression for the joint probability:

Lemma 2

If the joint probability distribution can be factorized as $P(X, Y, W) = f_1(X, W)f_2(Y, W)$, then we have $X \perp Y \mid W$.

1. Provide a formal mathematical proof for the aforementioned Lemma 2. (6 points)

2. Utilizing Lemma 2, rigorously prove that in the graphical model from Section IV, variable A is conditionally independent of variable D given variables B and C. (5 points)

V. Classical Multidimensional Scaling (MDS) (10 points)

Given an input dataset $X = \{X_1, \ldots, X_n\}$, write down the detailed algorithmic pseudocode for Classical Multidimensional Scaling (MDS).

VI. K-medoids Clustering Algorithm (10 points)

Provide the comprehensive pseudocode for the K-medoids clustering algorithm.

VII. Gaussian Mixture Model and EM Derivation (20 points)

For a given dataset $X = \{X_1, \ldots, X_n\}$ and a Gaussian Mixture Model with K components, write down the EM iterative update equations, including the E-step responsibilities and the M-step updates for the mixture weights, means, and covariance matrices.

VIII. Conceptual Deep Dives (10 points)

1. In the DBSCAN algorithm, is the designation of a 'core point' strictly deterministic when both the dataset and the hyperparameters are fixed? Provide a rigorous justification. (5 points)

2. Articulate the fundamental algorithmic and structural divergence between K-means and K-medoids clustering methodologies. (2 points)

3. When determining the optimal number of clusters (k), what is the exact mathematical or conceptual target objective of the elbow method? (3 points)